DATA ANALYSIS PROJECT NO -2

Language : Python

Task : Exploratory Data Analysis

IDE : VS Code

File Format : .ipynb

Findings :

1. 309 null values found in CouponCode column.
   Action : Replaced null values with “N/A”.
2. No duplicated values found in the dataset .

Analysis :

1. Performed **Descriptive Statisitics Summary** on column (Quantity, UnitPrice, ItemsInCart, TotalPrice).
2. Performed Outliers Investigation using IQR Method on column (TotalPrice)
   Result : **8 outliers** are found whose value is greater than normal order values.
   These outliers considered as premium customers instead of errors.
3. Mapped **Co-Relationship** between columns (Quantity, UnitPrice, ItemsInCart, TotalPrice)
   Method : Pearson correlation used.

Insights :

1. From Histogram graph, we see that lower-value orders make up the highest volume. Our business heavily depends on these smaller checkout amounts, while the demand for high-value total checkouts is low.

   Action : Focusing only on selling expensive premium packages won't work for most of our base. To make more money without spending extra on advertising, we should add simple push-promotions at checkout.

   Result : Showing alerts like "Add just $15 more to get free shipping" or suggesting small add-on items can easily turn a $400 order into a $600 order.

2. From Box Plot graph, we see that most of our regular orders fall between $500 and $1500.

   Action : Instead of offering generic discounts that lower our profits, we should introduce a "minimum spend" incentive. For example, we can offer free shipping or a small free gift only when an order crosses 1600. We can also suggest product bundles at checkout

   Result : This will encourage customers who are already spending $1000 to add one or two more small items to their cart, naturally pulling our average order values higher.